02LQD Specification and Simulation of Digital Systems
2022/09/16

1. Combinational circuit design (2 points)

Let X and Y be two BCD digits. Design in VHDL a **1-digit BCD adder** according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

It is up to the candidate to define the output signals and their size.

2. FSM design (16 points)

Design a FSM that serves as a controller for an elevator. The elevator can be at one of 3 floors: Ground, First or Second. There is one button that controls the elevator, and it has two values: Up or Down. Also, there are 3 lights in the elevator that indicate the current floor: Red for Ground, Green for First, Blue for Second. At each time step, the controller checks the current floor and current input, changes floors and lights in the obvious way.

1. Draw the state diagram of the Moore Machine (**4 points**)
2. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (**4 points**).
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 1 process (**4 points**)
4. Design the **Moore** FSM resorting in **Verilog** with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input `Addr` and a 1-bit input `Go` as well as the signals (to be defined by the candidate) required to access the RAM. When the `Go` signal is asserted for 1 clock cycle, the functional unit reads a memory address on its `Addr` input, then it computes the difference (positive or negative) between the number of bits at 1 and at 0 in the corresponding 16-bit cell in the RAM (*diffcount*). Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

  no later than Monday September 19 5 pm
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload ⇒ no correction ⇒ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.